GPT MEDICINAL OXYGEN 100% V/V

BRAND OR PRODUCT NAME

GPT Medicinal Oxygen 100% v/v.

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

Oxygen 100%v/v, Medicinal Gas, Compressed
There are no excipients.

PRODUCT DESCRIPTION

The drug product is gaseous oxygen medicinal, filled in cylinders as a compressed gas.
Gaseous oxygen is a colourless, odourless gas without taste.

The physical appearance of container is white colour cylinder shoulder and black colour cylinder body. The cylinders are filled with gaseous oxygen with pressure of 150 bar.

PHARMACODYNAMICS

Pharmacotherapeutic group: All other therapeutic products – medical gases, oxygen;
ATC code: V03AN01

Oxygen constitutes approximately 21% of air. Oxygen is vital for human life and must be supplied continually to all tissues in order to maintain the cells' energy production. Oxygen is transported in inhaled air via the airways to the lungs. In the pulmonary alveoli, as a result of the difference in partial pressure, gas exchange takes place from the inhaled air/gas mixture to the capillary blood. The oxygen is transported further in the systemic circulation, for the most part bound to haemoglobin, to capillary beds in the various tissues of the body. The oxygen is transported with the aid of the pressure gradient out to the various cells, its goal being the mitochondria in the individual cells, where it takes part in an enzymatic chain reaction, which creates energy.
By increasing the oxygen fraction in the inhaled air/gas mixture, the partial pressure gradient that controls the transport of oxygen to the cells increases.

Giving oxygen at a pressure higher than atmospheric pressure (HBO) considerably increases the amount of oxygen that is transported with the blood to the peripheral tissues. Intermittent hyperbaric oxygen therapy causes oxygen transport even within oedematous tissues and tissues with inadequate perfusion, and in this way can maintain cellular energy production and function.
In accordance with Boyle's law, HBO reduces the volume of gas bubbles in tissues in relation to the pressure with which it is given.
HBO counteracts the growth of anaerobic bacteria.

PHARMACOKINETICS

Inhaled oxygen is absorbed by a pressure-dependent gas exchange between alveolar gas and the capillary blood that passes the alveoli.

Oxygen is transported by the systemic circulation to all tissues in the body, mainly bound reversibly to haemoglobin. Only a very small fraction is freely dissolved in plasma. On passage through tissue, partial pressure-dependent transport of the oxygen to the individual cells takes place. Oxygen is a vital component in the intermediate metabolism of the cell. It is critical to the cell's metabolism, among other things, in order to create energy through the aerobic ATP production in the mitochondria.

Oxygen accelerates the release of carbon monoxide that is bound to haemoglobin, myoglobin and other iron containing proteins, and thus counteracts the negative blocking effects caused by carbon monoxide binding to iron.

Hyperbaric oxygen therapy further accelerates the release of carbon monoxide, compared with 100% oxygen under normal pressure.

Oxygen that is absorbed in the body is eliminated almost completely as carbon dioxide formed in the intermediate metabolism.

INDICATIONS
Normobaric oxygen
Medicinal Oxygen under normobaric pressure is used for:

- Treatment or prevention of acute or chronic hypoxemia, irrespective of genesis.
- As part of the fresh gas supply in anaesthesia or intensive care.
- As the driving gas in nebulisation therapy.
- As first aid treatment with 100% oxygen in decompression accidents, and suspected carbon monoxide poisoning.

The treatment is indicated in all age groups.

- Treatment of an acute attack in patients with an established diagnosis of cluster headache.

This treatment is indicated in adults only.

Hyperbaric oxygen (HBO)
Medicinal oxygen under hyperbaric pressure (HBO) is used for treatment of conditions where it is beneficial to increase the oxygen content of blood and other tissues above which is attainable under normobaric pressure.

- Treatment of decompression sickness, air/gas emboli of other genesis
- In carbon monoxide poisoning. HBO therapy is indicated essentially in patients who are or have been unconscious, that have shown neurological signs, cardiovascular dysfunction or severe acidosis and in pregnant females all irrespective of carbon monoxide haemoglobin (COHb) levels.
- As adjunctive treatment for osteo-radionecrosis and clostridial myonecrosis (gas gangrene).

The treatment can be used in all age groups.

RECOMMENDED DOSAGE
Normobaric oxygen

<u>General recommendations</u>
The primary purpose of oxygen therapy, i.e., correction of hypoxia, is to ensure that the partial arterial oxygen pressure (PaO_2) is not less than 8.0 kPa (60 mmHg) or that the oxygen saturation of haemoglobin in arterial blood (SaO_2) is not less than 90%. This is achieved by adjusting the fraction of oxygen in inspired gas. The lowest oxygen fraction of inspired gas needed to achieve the desired result of therapy i.e. a safe PaO_2 should be used. The therapy should be evaluated continuously, and the effect of treatment measured with PaO_2/SaO_2 or an estimate of SaO_2 i.e. SpO_2. The oxygen fraction of inspired gas should be adjusted according to each individual patient's unique requirement, taking the risk of oxygen toxicity into account (See Symptoms and Treatment of Overdose). In severe hypoxia, oxygen fractions that may involve a risk of oxygen toxicity can still be indicated.

<u>Acute or chronic hypoxia - Spontaneous breathing - Short term therapy</u>
In emergency medicine, oxygen is commonly administered by nasal prongs with at flowrate of 2-6 l/min or by face mask with a flow rate of 5-10 l/min. Patients, not at risk of respiratory failure and with an initial SpO_2 <85%, can be treated with 10-15 l/min by mask with a reservoir. Patients with known suspected chronic respiratory disease (e.g. COPD) that may have reduced chemoreceptor sensitivity should be treated with caution as a too liberal use of oxygen may cause respiratory depression. When 100% oxygen is indicated, a face mask with reservoir (oxygen flow should be enough to keep the reservoir partially or completely filled – i.e. not collapsed during - breathing) or a demand valve system should be used.

The fraction of oxygen in inspired gas should be kept so that with or without positive end-expiratory airway pressure (PEEP) or continuous positive airway pressure (CPAP) a PaO_2>8 kPa is maintained. The effect of short-term oxygen therapy should be monitored by repeated measurements of PaO_2 or by pulse-oximetry, which provides a numerical value for the SpO_2. However, these indices are only indirect measures of tissue oxygenation. Clinical assessment of the treatment is of outmost importance.

Acute or chronic hypoxia - Spontaneous breathing - Long-term therapy

In long-term therapy, oxygen can be administered by specially designed masks, e.g. Venturi-masks where the delivered oxygen concentration can be adjusted and dependent on the gas flow and the valve on the mask. Concentrations of 24 to 35% are commonly used.

The need for medicinal oxygen should be determined by obtaining arterial blood gas values and/or by monitoring SpO_2. The inspired oxygen should be titrated when used for long term oxygen therapy in patients with chronic hypoxic respiratory failure. A SaO_2/SpO_2 between 88 and 92% is commonly assessed as adequate in patients with chronic obstructive pulmonary disease (COPD). A too liberal administration can increase the oxygen SaO_2/SpO_2 clearly above the patient's normal range, which may cause respiratory depression because of chemoreceptor insensitivity for CO_2. Blood gases should be monitored to avoid excessive retention of CO_2 in patients with hypercapnia or reduced CO_2- sensitivity, in order to adjust the oxygen therapy.

Fresh gas supply in anaesthesia or intensive care - Assisted or controlled ventilation

Oxygen is commonly used in the intensive care setting. The inhaled oxygen should be titrated to the individual patient's need. The oxygen is commonly administered by assisted or controlled ventilation.
A PEEP is commonly applied to facilitate the ventilation/perfusion matching, recruiting airways and lung volumes, and thereby subsequently decreasing shunt.
During general anaesthesia a fraction of inspired oxygen about 0.3 is usually adequate. Higher fractions can be used in patients when found necessary.
If the oxygen is mixed with other gases, the oxygen fraction should be maintained at least at 0.21 in the inhaled gas mixture. The fraction of inspired oxygen can be increased up to 1.0.

Nebulisation

When oxygen is used for nebulisation, it can be used as a sole driving gas (100% in enough flow rate in order to nebulise the fluid in the nebulisation chamber) or mixed with air. In nebulisation therapy, the flow of oxygen and/or oxygen mixed with air is usually a continuous flow of 6-8 l/min.

First aid treatment

For emergencies if an acute administration of 100% oxygen is indicated, a face mask with reservoir (oxygen flow sufficient to keep the reservoir un-collapsed during breathing) or a demand valve system should be used.

Cluster Headache

Oxygen administration should be instituted as soon as possible after the onset of the attack. Oxygen should be delivered by facemask in a non-re-breathing system with a continuous flow of 6 to 12 l/min, for about 15 min.

Paediatric population

The safety and efficacy of oxygen in children of all ages is well established. The dosing instructions for the paediatric population are the same as for adults, except for new-borns (term, near term, and preterm). In newborns careful monitoring should be performed during the treatment. Oxygen in concentrations up to 100% can be administered in order to ascertain adequate oxygenation but for shortest time possible. Oxygen may be used during resuscitation in new-borns, but guidelines recommend that air is used initially. The lowest effective concentration should be sought in order to achieve an adequate oxygenation. Oxygen in low concentrations up to 40% (FiO_2 0.4) in combination with CPAP is recommended as an initial therapy.
There is no relevant use of oxygen in the paediatric population since oxygen for treatment of an acute attack in patients with an established diagnosis of cluster headache is not indicated in the paediatric. population.

Hyperbaric oxygen

General recommendations
Qualified healthcare professionals should administer HBO. HBO means that 100% oxygen is delivered at a pressure above the atmospheric pressure at sea level (1 atmosphere=101.3 kPa=760 mmHg). For safety reasons the pressure of HBO should not exceed 3 atmospheres.
The duration of a single treatment with HBO at a pressure of 2-3 atmospheres is normally between 60 min and 4-6 h depending on the indication. Sessions may, if necessary, be repeated 2-3 times a day depending on the indication and the patient's clinical condition. Compression and decompression should be slow in accordance with common routines in order to avoid the risk of pressure damage i.e. baro-trauma. The length and frequency of treatment should be decided by the attending physician taking the patient's physical and medicinal status into account. Recommendations for each indication are provided below.

Decompression sickness and air/gas embolism due to other reasons
HBO therapy with 2.5-3 atmospheres for up to 24 h, repeatedly as needed, is recommended.

Carbon monoxide poisoning
HBO therapy with 2.5-3 atmospheres is recommended. Usually, 45 min of treatment is required.

Osteo-radionecrosis and clostridial myonecrosis (gas gangrene)
For osteo-radionecrosis 2.4 atmospheres for about 90 min is recommended and for clostridial myonecrosis the recommendation is 3 atmospheres for about 90 min. The treatment could be repeated depending on therapy result.

Paediatric population
HBO can be used in children of any age. The length and frequency of treatment should be decided by the attending physician taking the patient's physical and disease status into account.

ROUTE OF ADMINSTRATION
Inhalation

CONTRAINDICATIONS
Normobaric oxygen
There is no absolute contraindication to oxygen therapy under normobaric conditions.

Hyperbaric oxygen
HBO is contraindicated in patients with an untreated pneumothorax, or other accidentally gas filled spaces with no ability to vent.

WARNINGS AND PRECAUTIONS
Normobaric oxygen
Whenever oxygen is used, the increased risk for spontaneous ignition should be taken into account. This risk is increased in procedures involving diathermy, and defibrillation/electro conversion therapy.

As a general rule, high concentrations of oxygen should only be administered for the minimum time required to achieve the desired clinical outcome. The inspired oxygen concentration should be reduced as soon as possible to the lowest concentrations needed. The patient should be monitored by repeated analyses of the PaO_2 or SpO_2 and the concentrations of inhaled oxygen should be titrated to maintain these parameters at an acceptable clinical level.

Prolonged exposure to higher concentrations of oxygen than listed below may generate oxygen species/free radicals and subsequently cause inflammation. Thus, the risk for oxygen induced lung dysfunction (e.g. signs or symptoms of acute lung injury/respiratory distress syndrome) shall be acknowledged.

The benefit versus risk for prolonged exposure to high concentrations shall be assessed on an individual basis. The evidence in the supporting literature suggests that the risk of oxygen toxicity can be minimised if the treatment follows these guidelines [fraction of inhaled oxygen in the inhaled air/gas mixture (FiO_2)]:

- Oxygen in concentrations up to 100% (FiO_2 1.0) should not be given for more than 6 h.
- Oxygen in concentrations of 60-70% (FiO_2 0.6-0.7) should not be given for more than 24 h.
- Any oxygen concentrations >40% (FiO_2>0.4) can potentially cause damage after 2 days.

In cases of high concentrations of oxygen in the inspiratory air/gas, the concentration/pressure of nitrogen is lowered. As a result, the nitrogen concentration in tissues and lung (alveoli) is lowered. If oxygen is taken up from alveoli to the blood faster than additional oxygen is delivered by ventilation, alveoli may collapse, causing atelectasis.

The formation of atelectatic lung areas may impair oxygenation of arterial blood because there will be no gas exchange in the atelectatic area despite perfusion. As a consequence, there will be a ventilation/perfusion mismatching and subsequently an increased shunt.

In patients with reduced sensitivity for carbon dioxide pressure in arterial blood, high concentrations of oxygen may cause, respiratory depression subsequently causing carbon dioxide retention, which in extreme cases can lead to carbon dioxide narcosis.

Paediatric population
Special caution should be taken when treating new-born since they have lower levels of defence systems and less active scavenging of free radicals than other populations. Thus, the potential negative effects of hyper-oxygenation are increased in the preterm and near term. The absolute lowest concentration, which gives the desired result, should be used in order to minimise the risk of ocular damage, retrolental fibroplasia (ROP), and broncho-pulmonary dysplasia (BPD) or other potential undesirable effects, which occur with a much lower oxygen concentration/fraction than in other populations.

Hyperbaric oxygen
Compression and decompression should be slow in order to avoid the risk of pressure damage i.e. baro-trauma.

HBO should be used with caution during pregnancy and in females of child bearing potential due to a potential risk of oxidative stress-induced damage in the foetus. The use should be evaluated in each individual patient.

HBO should be used with caution in patients presenting with pneumothorax or other accidentally gas filled spaces with no ability to vent (e.g. the pneumo-pericard) and who are treated with a chest-tube and/or patients with a medical history of pneumothorax. The use should be evaluated in each individual patient with regard to the risk of a new (tension) pneumothorax.

Use of greasy substances e.g. cosmetics should be avoided in order to minimise the risk for spontaneous ignition.

Paediatric population
The experience in new-borns, children and adolescents is limited. HBO should therefore be used with caution in the paediatric population. The benefit-risk should be evaluated in each individual patient.

INTERACTION WITH OTHER MEDICAMENTS
The pulmonary toxicity associated with high concentrations of oxygen (see Warnings and Precautions) can be exacerbated by concomitant use of anticancer drugs e.g. bleomycin, cisplatin, and doxorubicin, anti-arrhythmic drugs e.g. amiodarone, antibiotics e.g. furadantin (nitrofurantoin), alcohol deterrents e.g. disulfiram, and chemicals e.g. paraquat.

In the presence of oxygen, nitric oxide is rapidly oxidized to form superior nitrated derivatives that are irritant for the bronchial epithelium and the alveolocapillary membrane. Nitrogen dioxide (NO_2) is the principal compound formed. The oxidation rate is proportional to the initial concentrations of nitric oxide and oxygen in the inhaled air, and to the duration of contact between NO and O_2.

Paediatric population
There are no known other interactions than those in the adult population.

PREGNANCY AND LACTATION
Normobaric oxygen

No studies examining the potential toxicity of normobaric hyperoxia on embryo-foetal development or reproduction have been identified in the literature.

Pregnancy
Oxygen supplementation has no known negative effects on the foetus. Women of childbearing potential can use oxygen.

Lactation
Oxygen supplementation has no known negative effects on the breastfeeding child. Oxygen can be used during lactation.

Hyperbaric oxygen

Pregnancy
HBO should be used with caution during pregnancy and in females of childbearing potential due to a potential risk of oxidative stress-induced damage in the foetus. In severe carbon monoxide intoxication, the benefit versus risk for the use of HBO should be evaluated in each individual patient.

Lactation
There are no known adverse effects of HBO on lactation. However, lactation should be avoided during the actual HBO therapy.

SIDE EFFECTS
Summary of the safety profile
The most serious side effects that may occur are severe difficulty in breathing, so called respiratory distress syndrome. Too liberal oxygen administration may also cause respiratory depression in susceptible patients with reduced chemoreceptor sensitivity as seen in e.g. some patients with chronic obstructive pulmonary disease (COPD). It is not known how common this side effect is.

Paediatric population
In the use of oxygen in new-borns the risk for retrolental fibroplasia in pre-terms and the development of broncho-pulmonary dysplasia should be acknowledged. Except for these two risks, there are no other undesirable effects reported than in adults.

Tabulated summary of adverse reactions

Psychiatric disorders	
Not known:	**HBO**: Anxiety, confusion
Nervous System Disorders	
Not known:	**NBO and HBO**: Respiratory depression (chemo-receptor sensitivity) **HBO**: Loss of consciousness, epilepsy unspecified
Eye disorders	
Not known:	**NBO and HBO**: Retrolental fibroplasia in prematures **HBO**: Myopia

Ear and labyrinth disorders	
Not known:	**HBO**: Feeling of pressure in the middle ear, tympanic membrane rupture
Respiratory, thoracic and mediastinal disorders	
Not known:	**NBO and HBO**: Atelectasis, pleuritis, Respiratory distress syndrome, Pulmonary fibroses, Bronchopulmonary dysplasia **HBO**: Sinus squeeze
Injury, poisoning and procedural complications	
Not known:	**NBO and HBO:** Burns **HBO:** Baro-trauma

***HBO:** Hyperbaric oxygen **NBO:** Normobaric oxygen*

SYMPTOMS AND TREATMENT OF OVERDOSE

In oxygen intoxication there may be pulmonary symptoms of chest tightness, dry cough and pain on inspiration. Care must be taken where symptoms cannot present (e.g. intensive care) since the onset of objective evidence for pulmonary oxygen toxicity occurs late in its development.

The oxygen therapy should be reduced or, if possible, stopped, and symptomatic treatment should be started in order to maintain vital functions (e.g. artificial ventilation/assisted ventilation should be given if the patient shows signs of failing respiration).

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Medicinal oxygen has no or negligible influence on the ability to drive and use machines.

METHOD OF ADMINISTRATION

Precautions to be taken before handling or administering the medicinal product, see Special Precautions for Disposal and Other Handling.

Oxygen is administered with the inspiratory air. On exhalation the exhaled gas, with any oxygen excess, leaves the patient and is mixed with the surrounding air. Oxygen should be administered via special equipment.

Normobaric oxygen

<u>Spontaneous breathing</u>

There are many devices intended for the administration of oxygen in spontaneous breathing patients for example:

• **Low-flow systems**

The simplest systems, which deliver a mixture of oxygen to the inspiratory air, e.g., a system in which the oxygen is administered via a simple rotameter connected to a nasal catheter or facemask.

• **High-flow systems**

Systems designed to provide a gas mixture corresponding to the patient's entire inspiratory atmosphere. These systems are designed to deliver a fixed concentration of oxygen that is not influenced (diluted) by the surrounding air, e.g., Venturi masks with a fixed flow of oxygen in order to give a fixed concentration of oxygen in the inspired air.

• **Demand valve**

A demand valve system (i.e., a valve triggered by spontaneous ventilation) is system designed to deliver 100% oxygen without entrainment of ambient air, intended for short duration of administration by mask.

• **High Flow Nasal Cannula systems is also an option for spontaneously breathing patients.**

Assisted and controlled ventilation
When oxygen is administered by assisted or controlled ventilation, an oxygen air mixture is commonly used in order to achieve a desired oxygen fraction. The gas can be administered by mask, tracheal tube or tracheostomy.

Fresh gas flow during general anaesthesia
During anaesthesia, special anaesthesia equipment is used. Anaesthesia equipment commonly consists of a specially designed breathing circle intended for partial re-breathing. A circle system with a carbon dioxide absorber allowing a portion of the expired gas to be re-circled and inhaled again is often used.

Extracorporeal membrane oxygenation
Oxygen is usually administered via inhalation but can also be administered through a so-called oxygenator directly to the blood e.g. in conjunction with heart surgery (when a heart-lung machine is used) or in patients with severe therapy resistant hypoxia requiring extracorporeal membrane oxygenation/extracorporeal lung assist (ECMO/ECLA).

Hyperbaric oxygen
HBO is given in a specially constructed pressure chamber designed for HBO treatment in which pressures up to three times atmospheric pressure can be maintained. HBO can also be given via a very closely fitting facemask i.e. a hood that closes around the head, or through a tracheal tube.

INSTRUCTION OF USE

General precautions

- Do not smoke or use a naked flame in areas where medicinal gases are administered. High oxygen fraction facilitates ignition and explosive fire thus, any source that may case an ignition must not be used. Electronic cigarettes shall not be used or charged in the vicinity of a patient undergoing oxygen therapy or the oxygen source itself.
- Medicinal gases must only be used for medicinal purposes.
- Never place a mask or nasal prongs directly on textiles while treatment is being carried out since fabrics that become saturated with oxygen can be highly flammable and cause a risk of fire. If this should occur, thoroughly shake and air the textiles.
- Never use grease, oil or similar substances, even if the cylinder valve sticks or if the regulator is difficult to connect.
- Handle valves and devices belonging to them with clean and grease-free hands (i.e. no use of hand cream, etc.).
- Use of greasy substances e.g. hand-cream should be avoided in order to minimise the risk for spontaneous ignition in conjunction with oxygen under high pressure (e.g. HBO).
- In case of cleaning of cylinders or attached equipment, do not use combustible products and especially not oil-based material. In case of doubt, verify compatibility.
- Prior to any use, ensure sufficient quantity of product remains to allow completion of the planned administration.
- Only use standard devices designed for administration oxygen administration.
- When delivered from the manufacturer, the cylinders should have an intact tamper evident seal.

Preparation for use

- Remove the plastic cover from the valve before use.

The instructions below are applicable for cylinders where a separate pressure regulator shall be connected before use:

- Use only regulators designed for product.
- Check that the connection on the coupling or regulator is clean and that the connections are in good condition.
- Never use pliers to force regulators that are designed to be connected manually, as this can damage the connection.
- Check that the regulator is properly attached before opening the valve.
- Open the cylinder valve gently, at least half a turn.
- Check for leakage according to instructions that accompany the regulator.
- In the event of leakage, close the valve and disconnect the regulator. Mark the defective cylinder, keep it separate and return it to your supplier.

Use of cylinder

- Close the cylinder in the event of fire or when not in use.
- Ensure cylinders are secured to a suitable cylinder support in vertical position when in use, to prevent them from falling.
- For cylinders equipped with integrated valves, the user should be prepared to change the cylinder when the pressure gauge is in the yellow zone and change it when it enters the red zone.
- For cylinders that are not equipped with residual pressure valves, the cylinder valve should be closed when a small amount of gas remains in the cylinder (approx. 2 bars). It is important to leave a slight residual pressure in the cylinder in order to protect it from contamination.
- After use, the cylinder valve should be closed with normal force and the regulator or connection depressurised.

Transportation of cylinders

- When being transported, the cylinders must be secured to prevent them from falling.
- Larger cylinders should be transported with appropriate type of trolley. Particular attention should be paid to ensuring connected devices are not accidentally loosened.

Special precautions for disposal and other handling

The cylinder package must not be disposed but returned to the supplier.

STORAGE CONDITIONS

- Cylinders should be stored at temperatures below 50°C.
- Cylinders should be stored in a well-ventilated place and protect from sunlight.
- Cylinders should be stored under cover, kept dry and clean, kept free from oil and grease, free from flammable material.

SHELF LIFE

3 years.

DOSAGE FORM AND PACKAGING AVAILABLE

Dosage form

Medicinal gas, compressed.
Gaseous oxygen is a colourless, odourless gas without taste.

Packaging

The cylinders are made of Aluminium or Steel, equipped Pin-Index Valve or Bull nose Valve. The cylinder body is painted Black and the cylinder shoulder is painted White.

Cylinder size (L water capacity)	Valve type	Cylinder material	Settled Pressure at 27°C (1 bar)	Content (m^3 oxygen at 27°C, 1 bar)
2.3L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Aluminium	150	0.35
2.3L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	0.35
3.4L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Aluminium	150	0.5
3.4L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	0.5
5 L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Aluminium	150	0.7
5 L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	0.7
10L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Aluminium	150	1.4
10L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	1.4
25L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	3.4
40L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	6.4
45L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	6.8
47L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	7.2

50L	Pin Index ISO 407 or Bull Nose BS341 No.3 (Top Outlet)	Steel	150	8.0

Not all pack sizes may be marketed.

NAME AND ADDRESS OF PRODUCT REGISTRATION HOLDER / MANUFACTURER

Product Registration Holder:
Gas Pantai Timur Sdn Bhd (58757-T)
Lot 300733 & 300734, Kawasan Perindustrian Gopeng
31600 Gopeng, Perak Darul Ridzuan, Malaysia

Manufacturer:
Gas Pantai Timur Sdn Bhd (58757-T)
Lot 104-105, Kawasan Perindustrian Pengkalan Chepa Fasa 1
16100 Kelantan, Perak Darul Naim, Malaysia

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23 October 2024